

```
#include<stdio.h>

int A[100],size,top;

//push an element into stack

void push(int item)
{
    if(top==size-1)
        printf("\n Stack is full");
    else
    {
        top++;
        A[top]=item;
    }
}

//pop an element from the stack

void pop()
{
    if(top== -1)
        printf("\n Stack is empty");
```

```
    else
    {
        printf("\n The popped element is %d",A[top]);
        top--;
    }
}

// Display the content of the stack

void display()
{
    int i;
    if(top== -1)
        printf("\n The stack is empty");
    else
    {
        printf("\n The element in stack");
        for(i=0;i<=top;i++)
        {
            printf("\n %d" ,A[i]);
        }
        printf("<-top");
```

```
}  
}
```

```
// status of the stack
```

```
void status()
```

```
{  
    float free;  
    if(top==-1)  
        printf("Stack is empty");  
    else  
    {  
        printf("Stack top element is %d \n",A[top]);  
        if(top==size-1)  
            printf("\n stack is full");  
        else  
        {  
            free=(float)(size-top-1)*100/size;  
            printf("\n Free space=%f ",free);  
        }  
    }  
}
```

```
//Function for main()
```

```
void main()
```

```
{  
    int choice,n;  
    top=-1;  
    printf("\n Enter the size of stack :");  
    scanf("%d",&size);  
    do  
    {  
        // MENU driven program  
        printf(" \n MENU");  
        printf("\n 1.PUSH \t 2.POP \t 3.DISPLAY \t 4.STATUS \t 5.EXIT");  
        printf("\n Enter the choice :");  
        scanf(" %d",&choice);  
        switch(choice)  
        {  
            case 1 : printf("Enter the value to be pushed :");  
                scanf("%d",&n);  
                push(n);  
                display();  
            case 2 : printf("Enter the value to be popped :");  
                scanf("%d",&n);  
                pop();  
                display();  
            case 3 : printf("Enter the element to be displayed :");  
                scanf("%d",&n);  
                display();  
            case 4 : status();  
            case 5 : exit(0);  
        }  
    }  
}
```

```

        break;
case 2 : pop();
        display();
        break;
case 3 : display();
        break;
case 4 : status();
        break;
case 5 : break;
default: printf("\n Invaild choice!Please try again");
}
}while(choice!=5);
}

```

OUTPUT

Enter the size of stack :3

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :2

Stack is empty

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :1

Enter the value to be pushed :7

The element in stack

7<-top

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :1

Enter the value to be pushed :4

The element in stack

7

4<-top

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :1

Enter the value to be pushed :8

The element in stack

7

4

8<-top

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :1

Enter the value to be pushed :6

Stack is full

The element in stack

7

4

8<-top

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :2

The popped element is 8

The element in stack

7

4<-top

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :2

The popped element is 4

The element in stack

7<-top

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :1

Enter the value to be pushed :9

The element in stack

7

9<-top

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :2

The popped element is 9

The element in stack

7<-top

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :2

The popped element is 7

The stack is empty

MENU

1.PUSH 2.POP 3.DISPLAY 4.STATUS 5.EXIT

Enter the choice :5